

Ex no: I

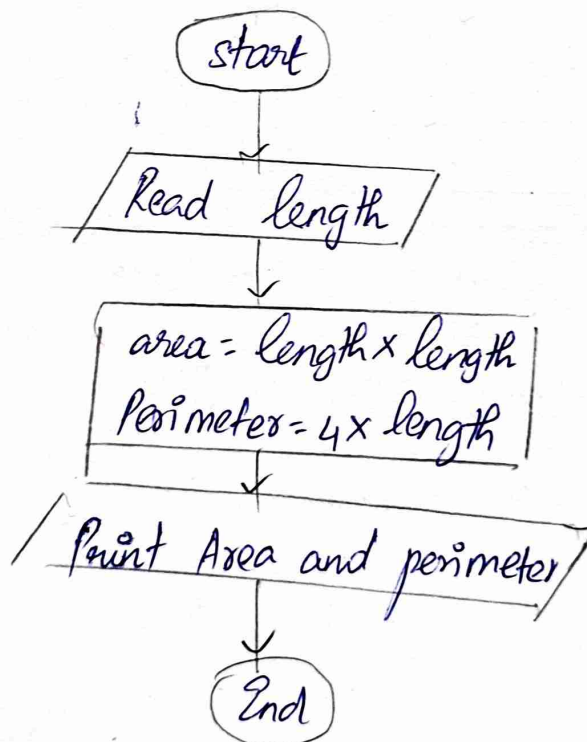
Date: 21/9/24

Calculate Area and Perimeter

Write an algorithm and draw a Flow chart to calculate the area and perimeter of a square
algorithm:

- step 1: Start the process
- step 2: Read the length
- step 3: Calculate area = $\text{length} \times \text{length}$
- step 4: Calculate perimeter = $4 \times \text{length}$
- step 5: Print area and perimeter
- step 6: End the process

Flow chart



Ex no: II

Days to Year Conversion

Write an algorithm and draw a flowchart to convert the given days into years & months.

algorithm.

step 1: Start the process

step 2: Input number of days

step 3: Calculate no. of years $\text{years} = \text{days} / 365$

step 4: Calculate remaining days $= \text{days} \% 365$ after years

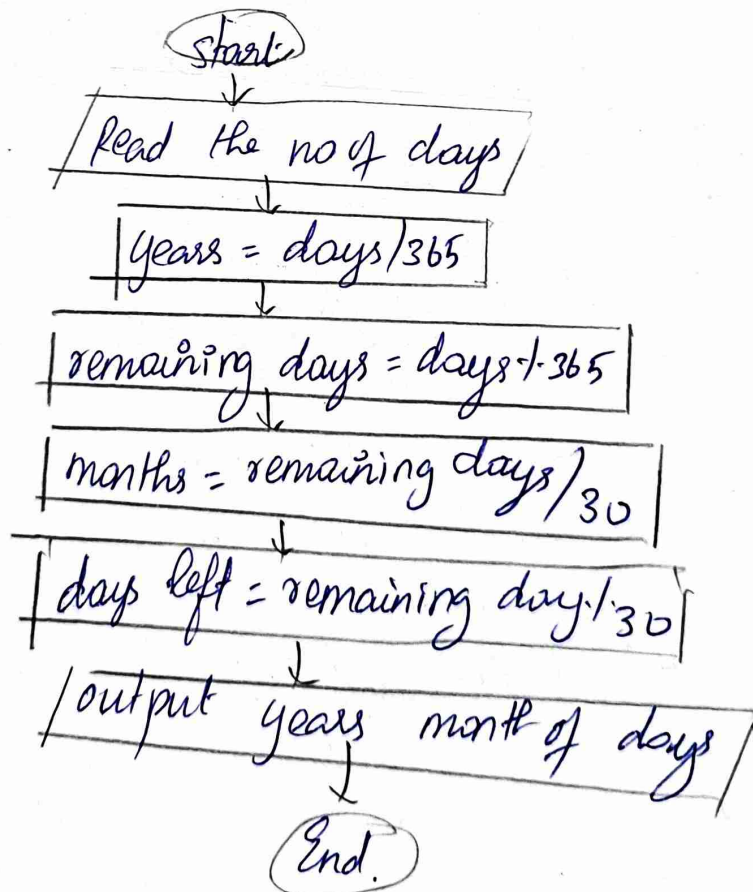
step 5: Calculate no. of months $= \text{remaining days} / 30$

step 6: Calculate remaining days left $= \text{remaining days} \% 30$ after months.

step 7: output the years, months and days left.

step 8: End the process

Flowchart.



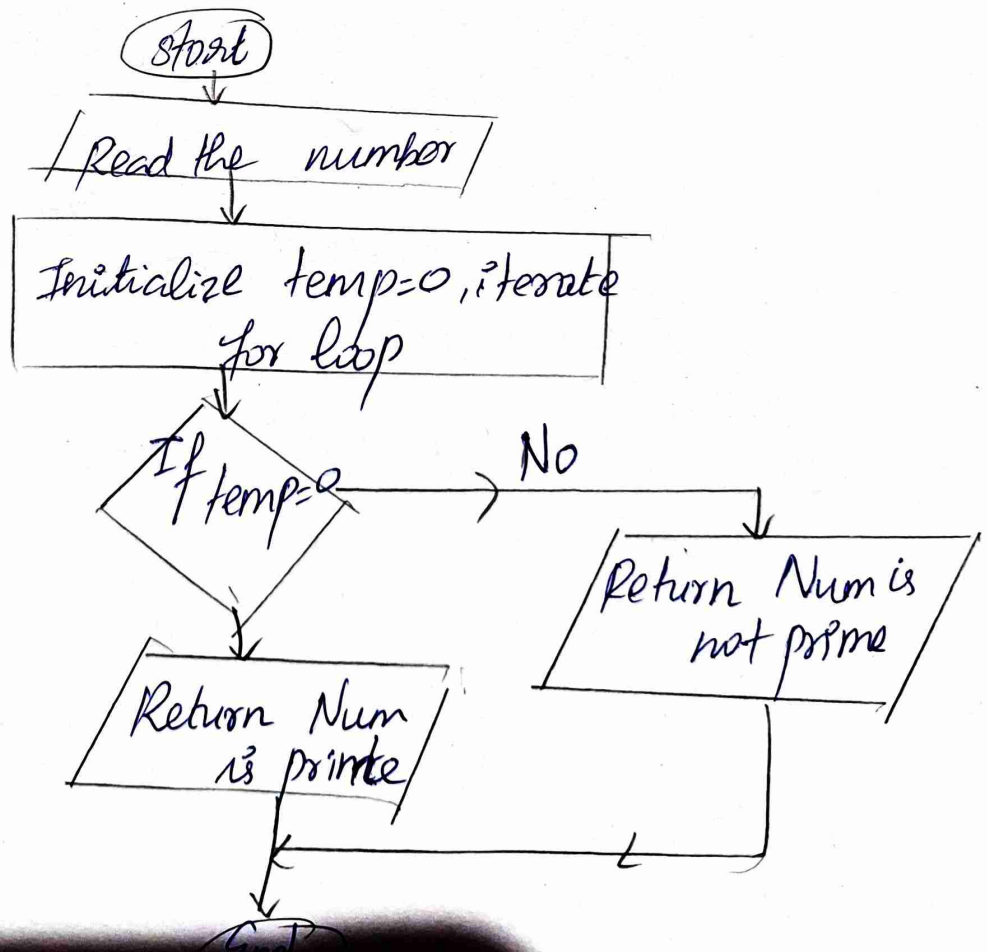
Prime number

Write an algorithm and draw a Flow chart to check whether the given number is Prime or not.

Algorithm:

- step 1: Start the process
 step 2: Take num as input
 step 3: Initialise temp = 0
 step 4: Iterate a for loop from 2 to num
 step 5: If num is divisible by loop iteration, the increment temp
 step 6: If the temp is equal to
 Return Num is prime else Return Num is not prime
 step 7: End the process

Flowchart:



Ex-no: IV

Leap year

241801328 - Yokiha

Date: 25/9/24

Write an algorithm and draw a flowchart to check whether the given year is Leap year or not.

algorithm:

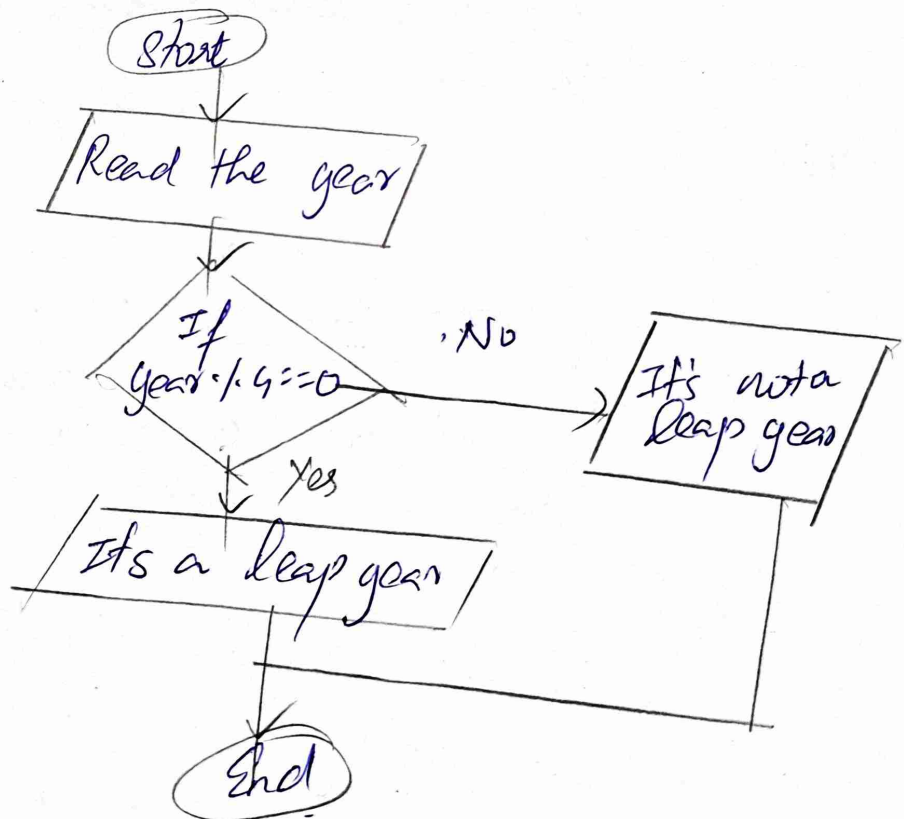
step 1: Start the process

Step 2: Read the year

Step 3: If the year $\div 4 = 0$ return "It's a leap year"
else return "It's not a leap year."

step 4: End the process.

Flowchart:



Ex. no: 5

241801327 - Yakiha

Date: 25/9/24

Palindrome Number

Write an algorithm and draw a Flow chart to check whether the given number is Palindrome number or not

Algorithm:

step 1: start the process

step 2: Read the number n

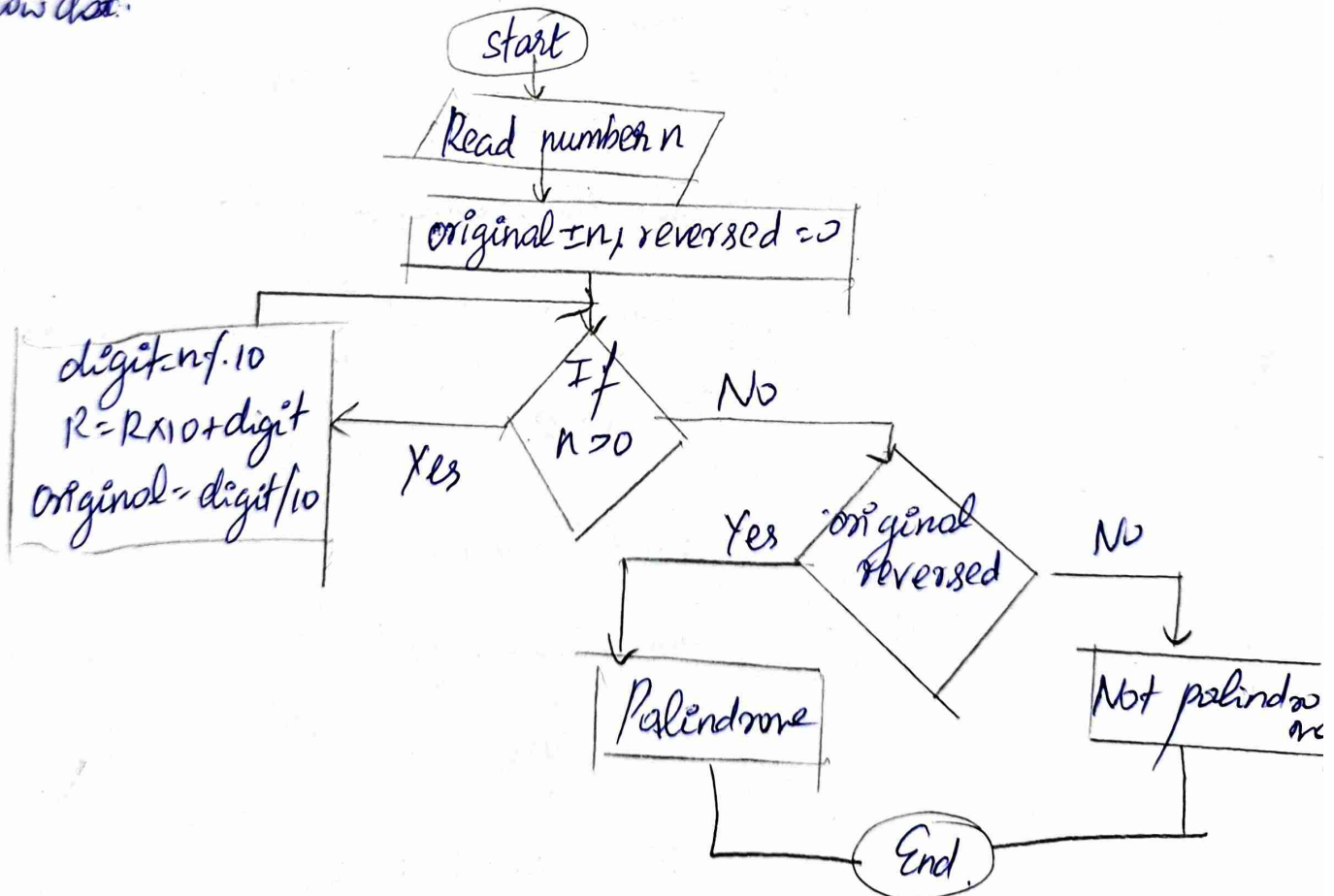
step 3: Initialise original $= n$ and reversed $= 0$

step 4: while $n > 0$, set $digit = n \bmod 10$, update reversed = reversed $\times 10 + digit$, update $n = n / 10$

step 5: If original $=$ reversed, print "Palindrome"
else print "Not Palindrome"

step 6: End the process

Flow chart:



Ex: no 17

24/801327 - Yokitha

Date - 25/9/24

Sum of digits

Write an algorithm and draw a flowchart to calculate the sum of digits in the given number

Algorithm:

Step 1: Start the process

Step 2: Input the number n

Step 3: Initialize $sum = 0$

Step 4: Repeat the following steps while $n > 0$

- Extract the last digit of n , $digit = n \% 10$

- Add the digits to $sum = sum + digit$

- Remove the last digit from n : $n = n / 10$

Step 5: Output the sum

Step 6: End the process

Flow chart:

